



The Dutch Historical Censuses: Harmonization and RDF

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Abstract

Historical censuses have an enormous potential for research. In order to fully use this potential harmonization of these censuses is essential. During the last decades enormous efforts have been undertaken in digitizing the published aggregated outcomes of the Dutch historical censuses (1795-1971). Although the accessibility has been improved enormously, we have to cope with hundreds of heterogeneous and disconnected Excel tables. As a result the census is still for the most part an untapped source of information. We describe the main harmonization challenges of the census and how we work towards one harmonized dataset. We propose a specific approach and model in creating an interlinked census dataset in the Semantic Web using the Resource Description Framework.

Key words: Historical Censuses, Social and Economic History, Historical Demography, Harmonization, Semantic web, RDF

Introduction

Censuses are taken regularly by governments throughout history to gain a better understanding of populations and their different characteristics such as size, age-structure, household compositions, occupations and other socio-demographic aspects. The Dutch government collected census information not only to get a view of the state of the nation, but (since 1850) also to facilitate the construction and updating of the population registers by the municipal authorities (Den Dulk and Van Maarseveen 1999). Although sometimes lagging behind social reality, historical censuses contain specific information about a nations *population characteristics* and *needs* at a given *time* in history, providing invaluable snapshots of the state of a nation (Higgs 1996). For the period before the 20th century, the census is one of the only large-scale historical statistical data sources on population characteristics which are not strongly distorted, providing comprehensive geographical coverage (Ruggles and Menard 1995).

The first integral enumeration in the Netherlands started in 1795 under the French influence during the Batavian Republic. It took over 35 years before the next general population census was organized (1829). This was based on the Royal Decree of 1828 which stated that the census should be taken every ten years. Due to more awareness and protest with regards to privacy matters, but also political and budgetary aspects the last ‘traditional’ door-to-door census was held in 1971.

Although a high non-response was feared, only 2.3% refused to be counted in one way or the other. The 1971 census marks the end of the traditional census in the Netherlands which in total covered 17 census years for almost two centuries (Den Dulk and Van Maarseveen 1999). The end of the traditional censuses have not exempted the Dutch government in its obligation to meet European regulations and to collect this type of information about its population. Currently the census is harvested digitally from the municipal registrars¹.

Unfortunately, because of the existence of the population registers from 1850 onwards, the original census forms (1850-1947) were *not preserved*. However, from the earlier censuses (1829 and 1839) about 50% of the micro data are still kept in local archives (Author 2012). For the last two census years, 1960 and 1971, the micro-results have been preserved on tape (Van Maarseveen and Doorn 2001). For the period 1850-1947 the results of the census are only preserved on *aggregated* level and published as *tabular* data in books. The number of volumes depends on the specific census year. Although these books have been one of the most consulted sources of statistics in the Netherlands and have become a valuable source of information for researchers, the use and accessibility of these books is quite problematic and therefore limited. Physical presence and cumbersome manual efforts were required in order to extract meaningful data from the census. In order to provide better access to and use of the census data

¹ Dynamic register of the population per municipality

major efforts have been taken in the digitization of the census, starting in 1996. From this year onwards the Dutch Statistics (CBS) and the institute Data Archiving and Networked Services (DANS), worked together in digitizing the books with the aggregative results of the censuses from 1795-1947 and to improve the accessibility of the 1960 and 1971 micro datasets. The first step in the digitization process was to scan the books and publish them as images in order to provide better access to the historical census data and also to preserve this material for the future. Information which previously was poorly accessible to researchers (e.g. via different university libraries, CBS, different institutions etc..) was now made available by way of internet and cd-roms. However, the images are very difficult to handle. A single table in the census can be represented by hundreds of images. The next step therefore was the shift from *medium* to *content* conversion in which the images were manually transcribed into Excel workbooks. During the transcription process the choice was made to represent the census tables in a source-oriented manner, meaning that the tables in Excel should resemble the tables in the books as closely as possible (including the presentation of the tables). As a result the researchers ended up with 2,249 Excel tables instead of an integrated harmonized dataset. These Excel tables are the basis for the next steps in the digitization process of the Dutch historical censuses and form the starting point for our harmonization efforts.

Although now digital and computer processable, the historical Dutch census is still not being used to its full potential. Besides the data representation limitations of having thousands of heterogeneous and unconnected Excel tables, a common prerequisite which bounds researchers to fully reap the potential of the census relates to the problem of data harmonization. The disconnected Excel tables present many different classification systems which need to be *standardized* to allow temporal comparisons. Next to structural problems we find all kind of inconsistencies in the structure of our tables, but also in the data itself as both source and digitization errors have been introduced at different stages.

[Insert Figure 1 here]

Figure 1 shows the various stages of the digitization process from the census forms to Excel tables, into a harmonized database. The last step will be done by using Semantic Web technologies such as, the Resource Description Framework (RDF). This article consists of three main parts. In the first part we will elaborate on the historical background and varying structures of the Dutch census dataset and the way it has been handled in the digitization process so far. In the second part we will go into the specific problems of harmonization that we have to address. The third part we presents our specific approach, proposing a solution for going from the Excel tables to an interlinked harmonized dataset. We will describe *how we converted* the Excel tables into RDF using a specific method

which is suitable for heterogonous tables with different hierarchies, *how we modelled* this using a three-tier architecture, separating the raw data from the annotations and harmonization layers and *how we harmonize* and *query* our dataset after this conversion to allow longitudinal comparisons.

Dutch Census Dataset

When referring to the published results of the Dutch historical census data over the years, we have to distinguish three main types of census data: population, occupation and housing. The population census spans the entire range of our historical census dataset whereas the occupational censuses were only published for the census of 1849, 1859 and 1889 onwards. Until 1930, information on the housing statistics of the Netherlands was published as part of the population census which contained some tables with ‘housing statistics’. The first official housing census was introduced in 1947 and was connected with the population census. In 1956, the housing shortage and need for data about the housing market called for a new housing census and it was conducted separately, independent of the population census. The last official housing census was held in 1971, again together with the population census.

As a first effort in both preserving and providing better access to the original census books, different digitization projects were undertaken by Dutch Statistics and DANS (Doorn et al. 2001; Van Maarseveen 2008). These projects concentrated on the digitization of the census books and resulted in around 22,000 images representing all tables with the census results. Although better accessible and preserved, the images as such are very difficult to handle. A single table from the original census books can be represented by hundreds of images which as such are also quite unreadable on a normal screen without having them enlarged six or seven times. The second step in the digitization of the census focused on *content conversion*. Accordingly, the images were converted into computer processable files, i.e. *Excel files*. Experiments with Optical Character Recognition (OCR) did not lead to satisfying results, as a result the entire conversion was more or less done in a manual way. The main problem was that the automatic OCR conversion still needed extensive manual input such as checking and correcting *next to* the cost of digitization itself (Doorn et al. 2001). Figures 2 and 3 show examples of respectively (a part of) an image of the census table and the corresponding Excel table after digitization. The images as well as the spreadsheets are downloadable from the website <http://www.volkstellingen.nl>.

[Insert Figure 2 here]

[Insert Figure 3 here]

The main principle in the conversion process from images to spreadsheets was to represent the source as closely as possible. This source-orientated process means that no efforts were undertaken to harmonize the data and structure of the census tables. Each Excel table applies to a certain year, specific region (municipality, province and national total) and specific census type, i.e. population, occupation or housing census. In total, the electronic historical Dutch census consists of 2,249 tables with aggregated data waiting to be aligned and harmonized in order to allow studies over time and space.

[Insert Table 1 here]

Table 1 shows the distribution of the number of *aggregated tables* and *annotations* per census year (1795-1956). These annotations have different types of meaning. Throughout all the census tables we find numerous annotations, made in different ways. They may refer to annotations made in the census themselves or provide suggestions for corrections that were made during the conversion process into Excel. Given the source oriented approach the original figures in the tables

were not changed (although we found examples that indeed the source was corrected). We find annotations sometimes as comments in a cell, in another sheet or even as replaced values. Most of the annotations in the census are textual (whether a comment or interpretation) and only a small number are actual corrections to the data (numerical). All in all we deal with 33,283 annotations in the Excel files of which it is not possible to distinguish in a formal way between changes from the source or 'new' ones. About 40% of the annotations are provided by the census of 1889 alone. As the process of annotating the census is still ongoing, and will continue in the future (the Excel files are still being checked manually for conversion errors), we create a bottom up standard classification system from the current annotations in the census and propose a specific model in RDF in order to organize the annotations and deal with future changes in a consistent manner (see example in RDF section).

Harmonization: Problems and Solutions

In general, the structure of a census is subject to change from year to year due to different systems or classifications used. Census bureaus are known for changing the structures and classifications of the census to reflect the information need at a given time, making it impossible to query the entire range of the census without proper data harmonization, especially for researchers studying longer periods of time².

When dealing with historical statistical sources, especially census data which have been collected throughout different periods in history, researchers recognize the need for *harmonization* across the different sources as a fundamental activity. Censuses which are collected and digitized over long periods of time have significant limitations and are hampered with evolving variables, structures, observation methods, processing methods and classification systems which make it difficult to fully reap the potential of the census (Van Maarseveen 2008; Ruggles and Mennard 1995; Putte and Miles 2005). The Dutch historical census is no different and shares many of these problems and even worse, provides mainly *aggregated data* in tabular form to work with. In this final sprint towards a harmonized census database, we need to overcome these challenges and enable the use of the census in a systematic and longitudinal way. In addition to the

² <https://usa.ipums.org/usa/chapter4/chapter4.shtml>

problems with the annotations discussed in the previous section, in the following sections we will describe the key challenges we face in the harmonization of the Dutch census: working with aggregated data, changing variables, creating variables, structural heterogeneity, inconsistencies and changing classifications.

Aggregated data

Statistical census data are typically presented on aggregated levels. This answers the information needs of the public, politicians, government etc... at given times. The specific harmonization challenge of the Dutch historical census relates to the fact that we mostly have *aggregated* data to work with. Although this type of data are not specific to the Dutch census (e.g. Sweden, Belgium, UK), we aim to harmonize the data across all the census years using primarily aggregated data, in comparison to current efforts which mainly focus on a *per year* harmonization using aggregated data.

Due to the lack of micro-data, harmonizing aggregated census data on a diachronic basis is hampered as it's not possible to simply build or rebuild a classification. Unlike many similar census harmonization efforts (see section: Comparable Studies) we cannot reconstruct the (classification) systems at a micro level to suit our needs. The harmonization therefore concentrates on two

problems: First of all we need to harmonize the variables and values over time and secondly we have to harmonize the totals from the several hierarchical layers in which the census results are published, e.g. it is clear that the national total of some specific variable is not always the same as the enumeration of the provincial totals of that variable. For example we find that the total sum of the number of inhabitants in all municipalities for a certain province, does not match with the totals of that province. The lack of micro data necessitates the use of a combination of statistical approaches with regards to harmonization of aggregated data. Considering this, we are constrained to provide higher level aggregations, creation of new variables, estimations, using averages and ratios, interpolations, imputations and other methods necessary to provide harmonized variables.

Changing variables

A general characteristic in the evolution of census variables is related to the level of detail. The evolution of the census variables can be described through an evolution tree where we identify mainly four different scenarios with regards to the changes the variables undergo. Although different scenarios exist in the evolution of the variables, we also find variables which stay more or less the same over time such as the total number of inhabitants per municipality, however with variations in labeling (including different spellings). A first scenario is the

introduction of new variables (*creation*) to reflect the changing information needs at a given time. In some cases we find that certain variables were merely used for specific census years and removed from later censuses, we refer to this as *extinction*. Other common scenarios are the *merging* and *differentiation* (splitting) of variables throughout the census. We encounter this often in the occupational census where occupations tend to merge and split due to innovation (i.e. specialization, differentiation etc.) but also with geographical variables such as municipalities which have changed significantly over the course of time in the Netherlands (Van der Meer and Boonstra 2006). For example, the composition of the municipality of Rotterdam changed significantly over time, having nine major changes between 1886 and 1941. Examples of changing variables across time can be found throughout the occupational census. During its life span the Dutch occupational census underwent several structural changes. Accordingly, we can identify three different subsystems within the occupational classification system of the historical censuses, i.e.: 1849-1859, 1889-1909 and 1920-1947. Starting with simple listings of occupational titles the systems and variables evolved significantly over there years presenting a greater level of detail in the consequent census years.

Creating variables

The meanings of variables and concepts in our dataset are subject to change from census to census. While in some cases it is simply different labeling of a variable, we also find variations in variables which are much more difficult to harmonize. When working with aggregated data, the creation of new variables is a common practice in the harmonization process. Problems such as changing age categories require different statistical methods based on estimations. As we deal with aggregated data we cannot simply construct ‘age ranges’ to allow temporal comparisons. New variables need to be constructed to make age categories which cover censuses over the years. Whereas in some cases simple addition could be sufficient, this is not always the case. For example in order to make the age ranges ‘14-18’ and ‘19-20’ comparable with ‘14-15’ and ‘16-20’, possible solutions could be by making a new category ‘14-20’ or by way of interpolation making three new categories ‘14-15’, ‘16-17’ and ‘18-20’. Although these ‘age ranges’ are artificial (constructed by way of estimation, interpolation, imputation etc.), we aim to provide users *different* variables allowing researchers to choose a harmonization which fits their needs best. In other words, we deliver different harmonized variables about the same data.

The same applies to the use of classification systems in harmonizing the census. As there is no one best solution we provide the user with different solutions on the same variable. In the case of the occupational census DANS already connected our dataset to external classification systems such as HISCO (Van Leeuwen et al,

2002) in an early stage. However, the level of detail in the Dutch occupational census is more fine grained compared to the HISCO classification and using only this system would result in loss of detail. Next to using these types of external classification systems it's necessary to also apply a bottom-up approach and use the classifications from the census itself to preserve the fine grained detail of the census.

Although in some cases we provide various variables with a great certainty, others are based on statistical computations. For example, in some years the population total is not given explicitly, however by adding the total males and females we can reconstruct the 'population total' variable without any doubt with regards to the validity of the harmonization. In other cases however we need to do calculations (estimates, interpolations, extrapolations, averages, imputations etc...) with the data in order to provide *at least one* harmonized view / answer. In some cases, simply adding up *or* dividing a category according to a certain ratio could suffice. This is for example the case of the diamond workers in the occupational census, which received their own category and were no longer counted with stonecutters after 1889, see figure 4. Accordingly, in this specific case we can provide two different harmonizations.

[Insert Figure 4 here]

Structural heterogeneity

During the digitization of the historical censuses the choice was made to apply a source oriented approach and represent the images from the census books as closely as possible. Consequently, another problem which hampers the harmonization of the Dutch census is related to the structural heterogeneity of the tables, even though the nature of the information is comparable. We therefore encounter not only changes in the naming and evolution of the variables, but also in the way they were presented, i.e. the structure (layout) of the tables. Although our tables share the same encoding and representation language, we still had to deal with how to model the different structures when moving towards one system. Each table in Excel has its own hierarchy of how variables and data are presented. When building a database out of these different structures and hierarchies it becomes very difficult / impractical to find an overall model which would cover the entire dataset (without losing any information). As we aim to provide several harmonizations for the same problem, we don't want to commit to a certain model when converting the tables and aim to preserve the original structures.

The dissimilar levels of detail and changing structures of the tables calls for a more integral and harmonized approach. We work in the lines of Esteve and Sobek (2010) and define census harmonization as the creation of unified,

consistent data series from dissimilar census data. The first step in this part of the harmonization process is to get rid of unnecessary complexity by converting the content of the 2249 Excel tables into a unified dataset in the form of a database system which can be queried. In a very straightforward approach we have departed from the Excel tables and converted our dataset to RDF and stored it in a RDF database system (called a Triplestore) which we aim to build on. More information on how we did this will be elaborated in the next section (see Census to RDF section).

Changing Classifications

Classifications systems are used in the census in order to categorize the various variables and put them into meaningful groups (Begthol 2010). Changes in the structure of the census and the evolution of the variables are also reflected in the different classifications systems used in the Dutch historical census. Radical changes in the classification systems and coding from one year to another make it difficult for researchers to utilize historical censuses (Meyer and Osborne 2005; Pineo et al. 1977; Ruggles and Menard 1995). There are different classifications systems for demographical variables, the housing system, religious denominations, occupational and geographical classifications in the census. The creation of *standard classification systems* for each of these (internal) systems is a

prerequisite in order to make temporal comparisons possible. The classifications which are most subject to change are those relating to occupation. Until the 1889 census a simple classification of occupations was used in the Dutch census which counted all occupations into relatively broad categories without making any distinction in the kind of enterprise. After this period the occupational classification system changed significantly with the introduction of the occupational census and recorded both the occupations as well as the kind of enterprise in which the individuals were working (van Maarseveen 2008). One of the features of this new classification was that it also made a systematic difference between different types of hierarchical positions within an occupation/branch. Another example of a classification which evolved significantly is religious denomination. While in some years there is a simple classification representing the most major religious denominations such as “Protestanten” in other years we have a very specific differentiation of religious types such as “Église National Suisse” or “Kwakers” (‘Quakers’).

Dealing with Inconsistencies

It will be clear that, changing variables and classification systems, the structural heterogeneity of the tables and the aggregated character of the data in itself will be a cause for major inconsistencies when making one system of the several

censuses. However, inconsistencies are also present throughout the different censuses as they were published. The process of converting the data in the original census books to Excel files has not only introduced new digitization errors but also replicated source errors without any provenance which makes it impossible to distinguish between the two. Even the original census books as kept in the libraries have handwritten changes to the data as numbers have been corrected. It seems that if these corrections were digitized into the Excel files by way of annotations. The same happened with published corrections and with established mistakes during data entry. Therefore, Inconsistencies are not only present in the structure of the Excel files but also in the values / numbers throughout the different censuses which is an indication of the data quality but also the changing nature of the census.

We use different scripts to manage these inconsistencies³. Several quality checks are provided to the user with regards to the quality of the data. For example, we use outlier detection, which displays observations that are numerically distant from the rest of the data. Conformance to Benford's Law (Benford 1938) tests whether the frequency distribution of leading digits of all retrieved population counts is the same as the width of grid-lines on a logarithmic scale. Censuses are well-known distributions expected to obey Benford's Law, and in the Dutch

³ <https://github.com/BLINDED> FOR REVIEW

census case the law is met with great accuracy. These different methods are applied to check the quality of the data and improve the inconsistencies.

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Harmonization and the RDF approach.

In order to use this untapped source of information on the population characteristics of the Netherlands for almost two centuries, it's obvious that we need to depart from Excel and move towards a database system which can be queried uniformly. In this context we apply a specific knowledge representation model from the semantic web, the Resource Description Framework (RDF), in order to move away from the Excel files and provide better access to and use of the historical censuses using Linked Data principles. Using RDF as a knowledge representation model, we create a *one on one* copy of the structure and contents of the Excel files in the form of a (graph) database and separate the harmonization process from the data itself (Authors 2012; Author 2004). We facilitate the different harmonization views / interpretations by applying a three-tier architecture in RDF where we separate the raw *data* from the *harmonization* and *annotations* in the census. Doing so, we also guarantee provenance and access to the original data in a source oriented approach which has always been a point of attention in the digitization of the Dutch historical census.

The Semantic Web and RDF

Envisioned in 2001, the Semantic Web (Berners-Lee et al. 2001) was conceived as an evolution of the original World Wide Web, which is built essentially on documents. Most of the contents of the Web are designed for humans to read, but not for computer programs to process meaningfully. Computer programs are able to parse the source code of Web pages to extract layout information and text, but there are no mechanisms to process their semantics (Authors 2014). The “semantic” Web enables the sharing of content from databases and other structured data sources which are not directly published on the Web. The Semantic Web “is not a separate Web but an extension of the current one, in which information is given well-defined meaning, better enabling computers and people to work in cooperation “ (Berners-Lee et al. 2001, 37).

More practically, the Semantic Web is also the collaborative movement and the set of standards that pursue the realization of this vision. The World Wide Web Consortium (W3C) is the leading international standards body and the Resource Description Framework⁴ (RDF) is the basic layer on which the Semantic Web is built. W3C defines RDF ‘as the standard model for data interchange on the Web and has features that facilitate data merging, specifically supporting the evolution

⁴ <http://www.w3.org/TR/rdf-primer/>

of schemas over time'⁵. It is used as a conceptual description method in computing. Entities of the world are represented with *subjects* and *objects* while the relationship between the two are represented with *predicates* connecting them (e.g. "Amsterdam" "is located in" "The Netherlands"). Hence, RDF is a knowledge representation system where facts and their properties are expressed as subject-predicate-object sentences known as *triples* (e.g. Amsterdam-isLocatedIn-TheNetherlands) and all connected have the form of a graph. Finally, all unique subjects, predicates and objects are assigned a Uniform Resource Identifier (URI) that uniquely identifies them on the Web. Once converted and published, RDF data can be queried online through the query language SPARQL⁶ (SPARQL Protocol and RDF Query Language).

There are several reasons why RDF is chosen as the data system in which we model, publish and query the Dutch census dataset. First, a graph data model like RDF is appropriate when the dataset suffers from structural heterogeneity. This is especially true in our case, where data spans two centuries and the schemas behind the tables changed substantially from one census to the other. In fact, we have 2,249 disconnected tables with different hierarchical structures which we aim to preserve. Second, the RDF model allows data publishers to easily link their datasets to other RDF datasets on the Web. The Dutch census case enables us to

⁵ <http://www.w3.org/RDF/>

⁶ <http://www.w3.org/TR/rdf-sparql-query/>

build a hub of socio-historical information, where census numbers and variables can be easily linked to historical classifications of occupations, municipalities, regions, labour strikes and religions, as well as other cross-domain datasets such as DBPedia⁷. With such a linked dataset, extended and enriched census information can be retrieved combining data from the linked sources, for instance: number of workers per occupation and year versus number of labour strikes per occupation, year and municipality or region.

Comparable studies

Over the past years different efforts have been undertaken using RDF technology for greater census utilization. The 2000 U.S Census (Tauberer 2007) was converted to RDF providing population statistics on various geographic levels. Although not historical and harmonized only for that specific census, it deals with the same challenges in finding an appropriate data model to represent the census data in RDF. In Canada, The Canadian health census uses Linked Open Data (LOD), based on RDF in order to provide greater access to and usage of the data and to promote greater interoperability which cannot be achieved with conventional data formats (Bukhari and Baker 2013). Using a scalable and interoperable format such as RDF aims to make their data reusable across

⁷ <http://dbpedia.org/>

different platforms. In another comparable approach, in the context of a national large scale project regarding the management of socio-demographic data in Greece, Irene Petrou et al. (2013) have applied LOD technologies to the Greek population census of 2011. A similar goal here is to publish ‘traditional’ datasets into RDF and allow easier access and use of the census by i.e. third parties. Its aim is to develop a platform within which the Greek census is converted, interlinked and available in a LOD format. The 2001 Spanish Census project is another advocate of applying LOD technologies such as RDF to the census and encouraging the development of the open government philosophy (Fernández et al 2011). Using micro-data from the 2001 population census Fernandez et. al (2011) propose a solution for converting the data into open formats allowing greater discoverability, accessibility and integration of the data which is a recurrent theme in all of the mentioned projects.

All these projects have merely harmonized within the domain of each census year (using micro data as a starting point) and focused mainly on publishing the data.

In the following we will explain how we have used RDF in a novel way and propose a *three-layer model* to harmonize the data over time.

A Three-tier Model

In order to deal with the challenges of the census we model our dataset in a three-tier architecture according to the multi-tier architecture principles where layers are logically separated. In our model the architecture consists of the *raw data* layer, the *annotations* layer, and the *harmonization* layer (see Figure 5). The dependencies between the layers are represented in the figure with directional arrows. An arrow from *A* to *B* means that structure and data from *A* needs to be linked to structure and data from *B*.

[Insert Figure 5 here]

We separate the data in this way for several reasons. First, the census source data contained in the *raw data* layer should be preserved, even if it contains errors, in order to be able to trace data provenance in the RDF system and to have a digital copy of the source. Second, as mentioned before the process of correcting the census data are currently an ongoing process and will be so in the future. In order to cope with the different type of annotations in our dataset, we have extracted, standardized and modeled the annotations according to a RDF annotation standard. *How* we deal with these corrections / annotations will be elaborated in the next section. Finally, harmonization is a dynamic process (consisting of different practices) that affects how raw data are interpreted, transformed and

presented, and it may need to be customized according to multiple research requirements. Moreover, due to the ambiguity of some harmonization practices we want to provide several solutions to a problem.

Raw Data Layer

The *raw data* layer consists of a *one on one* copy of the original Excel sheets (see figure 2). The 2,249 Excel tables with their different structures are stored in this layer in the form of RDF graphs. Since the data contained in a census table is statistical data, we have designed a data model around the central concept of the *table cell* (i.e. a data cell in our Excel tables), according to the W3C RDF Data Cube vocabulary⁸. RDF Data Cube is the *standard* for modeling and publishing multidimensional data, such as statistics, in the Semantic Web.

Although the *layout* and *structure* of the Excel tables differ significantly across our dataset they contain the same basic structure of three areas: cells containing the data as such and columns and row headers defining the data. Although easy for humans to spot where the numbers and variables are, we need to specify for each table where the columns, rows and content area *start* and *end*. This is done by way of so-called bounding boxes by which we define the table layout of the

⁸ <http://www.w3.org/TR/vocab-data-cube/>

raw data layer. This helps us to keep track of the different table structures and deal with structural heterogeneity.

Harmonization Layer

Harmonizing the Dutch historical censuses is based on a combination of different harmonization practices. On the one hand, we need to clean, correct, enrich, standardize and even restructure the data to have an *acceptable* dataset to do research with. All these ‘improvements’ to the data are part of the harmonization package and are sometimes even necessary before being able to continue in the harmonization process itself. We find for example spelling mistakes and variants, contents of columns which have shifted to another column or wrongly merged due to digitization errors, e.g. we find housing types under the municipality column, municipalities under the occupation columns etc. As no consistent logic is applied it is very difficult to extract the right data without extensive manual input.

Next to these types of harmonization practices we apply a combination of bottom up and top down approaches in order to further harmonize the census and make consistent classifications and variables. These include the creation of standard vocabularies, the construction of variables across the different censuses, creating new variables and values and connecting the variables to existing classification

systems such as HISCO (for occupational variables) and the Amsterdam Code (classification for municipalities in the Netherlands). For this connection we need to convert these classifications into RDF. Although, we use standard vocabularies whenever possible, the censuses often requires that we create our own classifications and vocabularies.

To maintain the detail of the data we need to create bottom up classification systems for the housing types, religious denominations, lower level classifications of occupations and of the municipalities such as neighborhoods. We store all these types of created data (mutations, new variables, classifications etc..) in the harmonization RDF layer which can be enriched and modified as a continuous iterative process without compromising the underlying data.

Three Cell Flag System

We have described harmonization as the process of creating a unified and consistent data series from various census tables. This process of *creating* requires interpretations (changes) to the data. In order to deal with these interpretations we have a data layer which we call the *Three Cell Flag System*. This is the nucleus of our approach and means that we have three variables for each cell-value of the census, namely; the original value, the interpretation (which may be the original or a new value) and a flag, indicating the nature of the interpretation. For example, if a cell has the original value of '39', and cross validation showed that

‘39’ was a typo and should be ‘93’, the interpretation gets the correct value ‘93’ and the flag will indicate that the corrected value was based on cross validation over the row and column totals. Moreover, we also have a flag indicating that we accept the original value as correct, resulting in the same values for the interpretation and the original.

When we combine two variables to create a higher level aggregation for harmonization purposes we in fact create a new level of the *Three Cell Flag System*. Building on the example illustrated in figure 4, where we harmonized the ‘stonecutters’ and ‘diamond workers’ of 1899 into one group to make it comparable with the census of 1889, we combine the values of both groups of 1899 into one and the same value both for the original value and the interpretation. The interpreted value may be further modified in this action, indicated by a different flag value. The other way round: splitting a value of one group into values for two subgroups is different in that we have no original to transfer, but immediately interpolate to achieve two interpreted values where the flag indicates the rule we have used to split the original value.

Harmonizing in Four Stages

Practically, we harmonize the dataset in four stages. Firstly, we define which standard variables and values we will use for the raw variables and values which we find in the censuses (in RDF Data Cube terminology respectively *dimensions*

and *codes*). We try to use as much as possible existing standardized variables, e.g. the already mentioned Amsterdam code for Dutch municipalities (Van der Meer & Boonstra 2006). Secondly, we map the original variables and values of the census tables with the standard ones. Thirdly, we define transformations to create new variables or values, e.g. the creation of new age categories or to split into occupational (sub)classes (see figure 4). Fourthly, we use the outcomes of all previous stages to *smooth* the data into one system, harmonizing the totals of all kind of geographical layers with each other to achieve a consistent outcome, e.g. the sum of all provinces on a specific variable should be equal to the national totals for all variables. Finally, bringing together the richness of our model we provide one big flat table for expert users as well as different harmonizations of the data which can be queried and linked to other datasets online.

Annotations Layer

Throughout their lifespan the censuses have been annotated in different ways, applying no consistent system, logic or provenance to how and why the annotations were made. Scattered and even sometimes hidden in different tables, we encounter annotations which were *source made* (e.g. annotations in the original books to note that females were included with the male population instead of having the usual separate column for females), *made during data entry*

(e.g. annotating that some specific figure could be wrong) and *corrections made after data entry* (e.g. correcting probable mistakes based on existing annotations or new found problems). Moreover, the way these different type of corrections were implemented differs greatly across the tables and census years. We find annotations which were made as cell comments in Excel, as notes or even in a separate Excel sheet with a reference to the changed value in a cell.

Because of this lack of structure and predictability, we cannot handle them as raw data and beforehand we extract all annotations from the Excel tables, standardize and model them in the annotation layer of our three tier model, using the W3C Community Open Annotation Core Data Model standard⁹. The created annotation layer is linked with the raw data layer (see Figure 5). For provenance purposes we also attach an author to each annotation. We flag the contents of the annotation to indicate the specific issue of this annotation using a second system of the aforementioned Flag System. Table 2 gives an illustration of the flagging of the most common annotations. Information of these annotations can be used to make interpretations in the harmonization layer and will be flagged with a content that refers to the used annotation.

[Insert Table 2 here]

⁹ <http://www.openannotation.org/spec/core/>

Census to RDF

Presently we are so far that we have converted all raw datasets from the censuses into an RDF triple dataset. Apart from all kind of experiments with linked data principles we have also harmonized a first subset of the data from the Population Census of 1859-1889. The first stage in moving towards a harmonized database from the Excel sheets begins with the conversion of the tables into RDF, using a script called TabLinker¹⁰ (Authors 2012). This script converts the structures of the tables into an RDF graph for each census table (in Excel).

As noted earlier, we need information about the layout of these tables in order to create a one on one version of the tables in RDF. In principle a table consists of three main regions: *columns*, *rows* and the *content area* with the actual numbers. This content area needs to be linked with the relevant column and row headers. Although the structure and hierarchies in the tables differ greatly from table to table, they all share common regions with regards to the layout. Exploiting this common characteristic allows us apply the same approach in converting all tables to RDF. To maintain the structures of the tables TabLinker needs to define different styles to link all cell values to the corresponding columns and rows. Using standard functionalities in Excel, we *color/style* the boxes of our data manually defining the columns, rows and cell values of each table (see Figure 6).

¹⁰ <https://github.com/Data2Semantics/TabLinker>

[Insert Figure 6 here]

This coloring process is very straightforward and creates a faithful (one to one) representation of the tables in RDF. *TabLinker* defines the following styles:

- **Title** marks cells that contain the table title and description, placed at the top left of the tables (this style is transparent and illustrated by a checkered version in figure 6).
- **Data** marks the data cells with the actual census numbers (the white colored section in figure 6).
- **ColHeader** marks the column headers of the table just above the content of the cells (the light blue colored section in figure 6). These headers contain the values for different variables such as: age ranges, marital status, sex, etc.
- **RowHeader** marks cells with row headers, usually placed at the left of the table (the caramel colored section in figure 6). These cells usually contain values for geographical variables like municipality.
- **HRowHeader** marks cells with hierarchical row headers (the purple colored section in figure 6). This style is similar to row headers, with the difference that these cells form hierarchies or taxonomies (e.g. occupations of class *I*, subclass *a*, group *Diamond workers*).

- **RowProperty** marks cells with the names of the row variables, placed at the upper left of the table (the dark blue colored section in figure 6). The column variables are usually not made explicit in the censuses (the values speak for themselves).
- **Metadata** are used to mark any additional defining data that the tables may contain, like references to column or page numbers of the census books (the orange colored section in figure 6).

For each census table we generate three RDF large interconnected graph-systems, shown as layers in Figure 5. Samples of such separate RDF graphs are shown in Figures 7 and 8. Figure 7 shows what TabLinker produces in the *raw data* layer (see Figure 5) for *one single data cell*, represented by the central circular node labeled `.x`. This node represents a specific cell of the census tables and its entire environment, namely: the column and row headers that defines the cell, the data contained in the cell, etc. In this case, the cell contains the value '1' (`"1"^^xsd:int`), and the headers defines the content of the cell as persons of 14 or 15 years old (`:14--15_1875-1874`), being a man (`:M`), being a single (`:O`) and working as roof tile maker (`Sheet1:I/E/Fabricage_van_dakpannen__pannenbakkers`), which is an occupation into the work category I (`:I`) and subcategory E (`:I/E`). As this is only a description

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for *one* single data cell in the table, having thousands of cells per table generates a much larger interconnected graph.

[Insert Figure 7 here]

Similarly, the sample shown in Figure 8 specifies an annotation (labeled :y) pointing to a cell with coordinates *E663* in the file *VT_1859_01_H1* and the province table *Noordbrabant*. It includes some metadata, such as the creation date of the annotation (June 21st, 2012), who generated it (somebody represented with the node *TOM*), and a link to the original value (central node in the raw data layer) labeled :x. It also contains the flag *1* (see table 2), which indicates that the annotated cell contains an incorrect value (*1*) that should read *10* (table 2).

[Insert Figure 8 here]

All these graphs are stored in an RDF database called *Triplestore*. From these Triplestore we will build the harmonized database to be distributed to researchers and we will provide access for live online querying via a SPARQL endpoint¹¹. With such an endpoint, users and applications can send census queries in SPARQL to a server holding the dataset, and retrieve results in multiple formats such as CSV, HTML, or others.

¹¹ <http://BLINDED> FOR REVIEW

Using the RDF data

To date the historical censuses were merely available in the form of Excel files. Researchers interested in longitudinal analysis, were forced to go through all 17 census years and collect all the necessary tables *before* doing any harmonization or analysis on the data. Exposing the generated RDF data on the Web via an SPARQL endpoint enables users and web applications to retrieve, analyze, and visualize the historical census data, which is now available in one system. So far, we have written client applications that query the endpoint and draw maps displaying the population of the Netherlands according to some user constraints (Authors 2012). We have developed query templates that allow us to systematically access the dataset in a homogeneous way. All Dutch historical census data (in our raw data layer) is now available in our SPARQL endpoint in one single Web address¹², exposing:

- 110,585,567 total triples
- 10,272,862 marked cells triples
- 389,132 hierarchical row headers
- 7,960,911 data cells
- 61,110 column headers
- 3,609 row properties

¹² See <http://Blinded> for review

- 2,150 titles
- 1,581,546 row headers
- 274,404 metadata cells

We have already seen some examples of how these layers interact (e.g. how an annotation influences the result retrieved from the query over a data cell). More importantly, at this point the three layers come together to give the census data all its expressive power. Client applications (i.e. applications that use such endpoint as a data source) can be developed independently by different types of users. The SPARQL endpoint can be seen, in fact, as an online database plug that any application can *leverage* via the Web. This gives researchers, historians and developers the opportunity to build their own applications on top of these data, which until now was only available via 2249 disconnected Excel tables.

Although some knowledge of SPARQL is needed to perform queries on these data, both humans and computers are currently capable to retrieve any piece of information of the Dutch censuses in the raw data layer (see Figure 5). Moreover, the raw data provides usefulness for *debugging* and *harmonization* purposes. Practically, querying the raw data enables us to extract the needed variables, assess the quality of the data, identify already common variables across the years for classification purposes, visualize them, detect outliers (for example visualizing

the population of Noord Brabant on the map revealed several cities, clearly falling outside of this province¹³), extract the concept drift across the years, identify where a variable breaks down over time for harmonization purposes, etc. By querying for a certain variable (e.g. an occupation, population size, municipality etc.) across the raw data we are able to see for which years this variable is present. We visualize this in a simple graph and identify the evolution of the variable across our dataset. Using these practices we can already construct the basic branches of our evolution tree, i.e. we can identify variable creation and extinction which could serve for the mapping of common variables across the years.

The enriching of the annotation and harmonization layers is an ongoing process. The first step towards an interlinked dataset has been set by exposing a raw copy of our data. Even though all the raw data are now exposed via RDF and available online, it still carries the same challenges when using it. To fully reap the potential of this dataset and allow comparisons over time it is crucial to keep building on the harmonization layer. By applying a combination of bottom up and top down approaches we have already enriched the harmonization layer with spelling variants, standard variables and values, external classifications system such as HISCO and the Amsterdam Code, bottom up historical housing classification etc.

¹³ <http://Blinded> for review

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Conclusions and further work

Censuses tend to represent social reality in a very specific way. They are susceptible to change in order to meet the information needs of a specific government or society, providing a contemporaneous view on societal reality. Harmonization of historical censuses is a prerequisite for fully reaping the potential of census data in scientific research. In the Netherlands census data have been collected for almost two centuries. Over the past twenty years various digitization efforts have been conducted which have left us with over two thousand heterogeneous Excel tables with aggregated census data, each with its own specific data structure. These digitized tables are our point of take-off in the harmonization process. The aggregated nature of our data leads us to an approach which is different compared to other census harmonization efforts such as IPUMS. We described the challenges associated with harmonization of aggregated historical census data, identified different harmonization types (practices) and proposed possible solutions in order to deal with i.e. changing classifications, the creation of variables based on aggregated data, structural heterogeneity etc. We applied a specific method in RDF to model our data with all their complexity. The changing structure of the census and ambiguity of the variables requires a design which is flexible enough to allow different harmonizations of the data and yet able to account for changes made during the harmonization process. To accompany this, we apply a Three Cell Flag system

which takes into account the original value of the data, the interpretations we assign to the variables, whether harmonized or not, and the specification of the actions which have been undertaken to harmonize or correct the original data.

After years of numerous efforts we are now on the verge of providing a much better access to the historical Dutch censuses. Currently, the digitized historical censuses are not only published in the Semantic Web but also available in one system for the first time since the digitization efforts. With some SPARQL knowledge, users are already enabled to query the entire census data contained in our raw data layer. Although this is the first step towards a harmonized dataset, coming from 2,249 excel files, researchers currently enjoy the benefits of having everything in one system.

In order to achieve this, we first had to transfer the Dutch Historical census data into RDF and model it according to current standards of the Semantic Web. We have developed a specific tool which deals with heterogeneous excel files (containing aggregated data) allowing us to convert the census tables into RDF in a very straightforward process, without losing any information from the original tables. As more images will be transcribed into excel tables in the future, this tool can easily be reused to expand the current dataset. Acknowledging the problems of the census we have designed a specific model in RDF where we separate the raw data from the annotations and harmonization. We have developed standard

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6 templates and interfaces for querying the data in a uniform manner and
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8 experimented with different visualizations to explore our dataset. Through an
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10 online interface which directly plugs into our raw data layer, we have already
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12 started with the standardization of variables and values and connected them across
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14 the years. Moreover, we were able to connect the municipalities and occupation in
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16 the historical censuses with existing classifications like HISCO and the
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18 Amsterdam code for Dutch municipalities. We have also developed new
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20 standards, i.e. for the different housing types, annotations and religious
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22 denominations. In order to test and validate our methods we have applied these
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24 practices on a subset of the Dutch historical population census (1859-1889) and
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26 were able to harmonize and query these variables across time. Building on these
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28 practices and the principle of our proposed three tier model we are continuously
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30 enriching the harmonization and annotation layers to allow further temporal
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32 comparisons.
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40 In contrast to current census harmonization efforts dealing with aggregated data
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42 (whether in RDF or with traditional approaches), we provide a model and specific
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44 approach for harmonizing historical censuses *across time*. By already exposing
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46 our raw data, we allow third party users to build their own datasets, harmonization
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48 and or tools on top of the data. Future work will consist of further enriching the
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50 harmonization layer of our model by creating and adding our own (bottom-up)
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52 harmonization vocabularies next to connecting to spin-offs of existing systems
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(such as HISCO2rdf). We aim to provide a web based user interface which allows the users to query the data based on variable selection and time series and export these data in different formats. Although we provide all the benefits of RDF we also consciously want to shield users from the RDF output and provide clean, *known formats* such as CSV, Relational Databases or even round-tripping to Excel tables so users can integrate these data into their own workflows.

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Figure 1 - Census digitization process

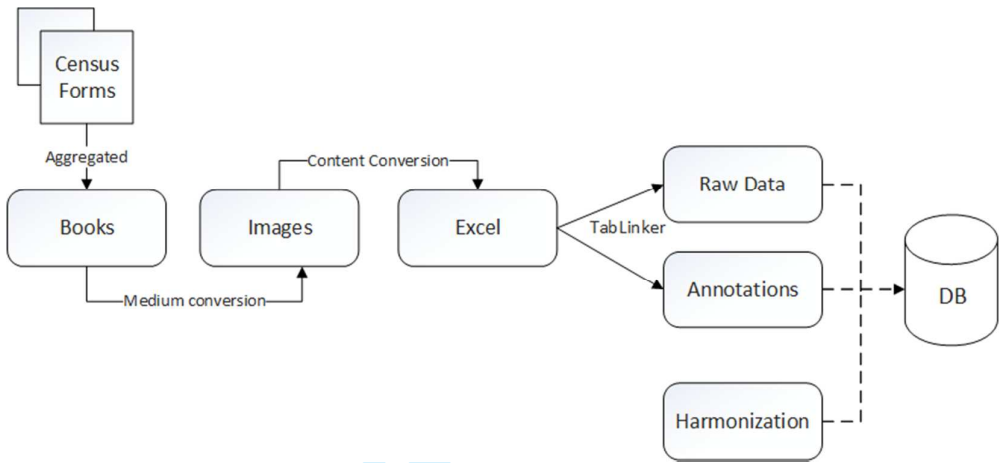


Figure 2 – Example of a scanned table image

EERSTE

PROVINCIE NOORDBRABANT.

GEMEENTEN. (Communes.)	PLAATSELIJKE INDEELING. (Divisions de communes.)	Woningen in de gemeente. (Demeures dans la commune.)				BEVOLKING. (Population.)								Tijdelijk (Temporaire- ment aanwezig. présente.)		
		Woonhuizen. (Maisons.)				Bij Présente de telling le jour du aanwezig. 'recensement.	Tijdelijk (Temporaire- ment afwezig. absente.)		TOTAAL. (Total.)							
		Bewoond. (habitées.)	Onbewoond. (non habitées.)	In aanbouw. (en construction.)	Bewoonde schepen. (Navires habités.)		M.	V.	M.	V.	M.	V.	M.	V.		
Aalst.	Kom. Kerkeind	68	4	"	"	"	133	133	"	"	133	133	2	"		
	Kerkeind	10	"	"	"	"	18	19	"	"	18	19	1	"		
	Buiten Achtereind	8	"	"	"	"	21	16	"	"	21	16	"	"		
	de kom. Ekenrooi	14	"	1	"	"	47	48	1	"	48	48	"	"		
	Laareind	35	5	"	"	"	80	65	"	"	80	65	2	"		
	Totaal binnen de kom	68	4	"	"	"	133	133	"	"	133	133	2	"		
	» buiten » »	67	5	1	"	"	166	143	1	"	167	143	3	"		
	Totaal	130	9	1	"	"	290	276	1	"	300	276	5	"		
Kom.	Wijk A. Huizen	23	"	"	"	"	56	50	"	"	56	50	1	1		
	» B. Instituut	1	"	"	"	"	"	90	"	3	"	93	"	"		
	Overige huizen	62	7	"	"	"	130	131	2	"	132	131	3	"		
	» A. Het Laar	9	"	"	"	"	25	17	"	"	25	17	"	"		
	» »	76	8	"	"	"	144	130	1	"	145	130	1	"		

Figure 3 – Example of a table converted to Excel from images

A	B	C	D	E	F	G	H	I	J	K	L	M
1	PROVINCIE NOORDBRABANT: EERSTE GEDEELTE: huizen, bewoond en onbewoond, bewoonde schepen en getelde pers											
2												
3	GEMEENTEN	PLAATSELIJKE INDELING					Woningen in de gemeente			Bewoonde schepen	Tijdelijk aanwezige schepen	Bij de aanvang van de M.
4							Woonhuizen					
		Kom/Buiten de kom	Wijk	Soort plaats	Naam	Onderkomens	Bewoond	Onbewoond	In aanbouw			
5												
6												
7	Aalst	Kom			Kerkeind		63	4				13:
8		BK			Kerkeind		10					1:
9					Achtereind		8					2:
10					Ekenrooi		14		1			4:
11					Laareind		35	5				8:
12		TK					63	4	0	0	0	13:
13		TB					67	5	1	0	0	16:
14		TOT					130	9	1	0	0	29:
	Aarle - Bistel											
15		Kom	A			Huizen	23					5:
16			B			Instituut	1					
17						Overige Huizen	62	7				13:
18		BK	A		Het Laar		9					2:

Figure 4 – Splitting of an occupational class

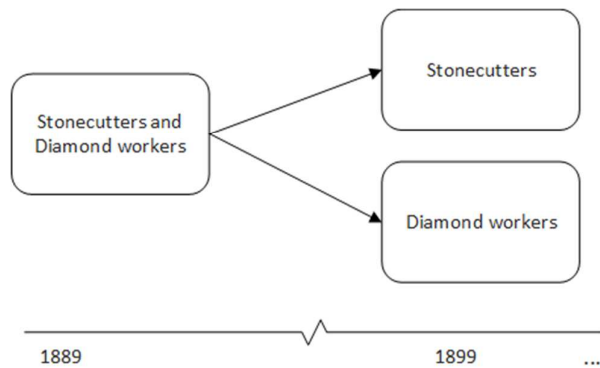


Figure 5 – Harmonization Three tier model

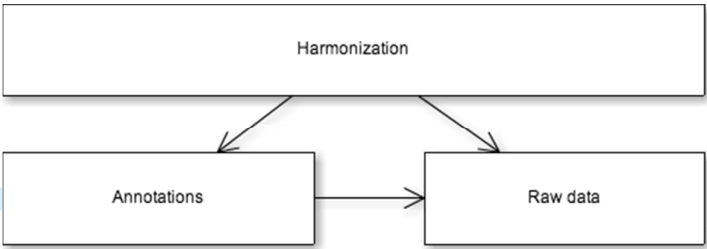


Figure 6 – Marked census table with TabLinker

RowProperty	HRowHeader	ColHeader	RowHeader	Data	Metadata	Title					
A	B	C	D	E	F	G	H	I	J	K	L
Tabel 1. Indeling der werkelijke bevolking naar de beroepen onder vijf en dertig beroepsklassen, gerangschikt in alphabetische volgorde; positie in het beroep, ge											
		</									

Figure 7 – Raw data layer graph

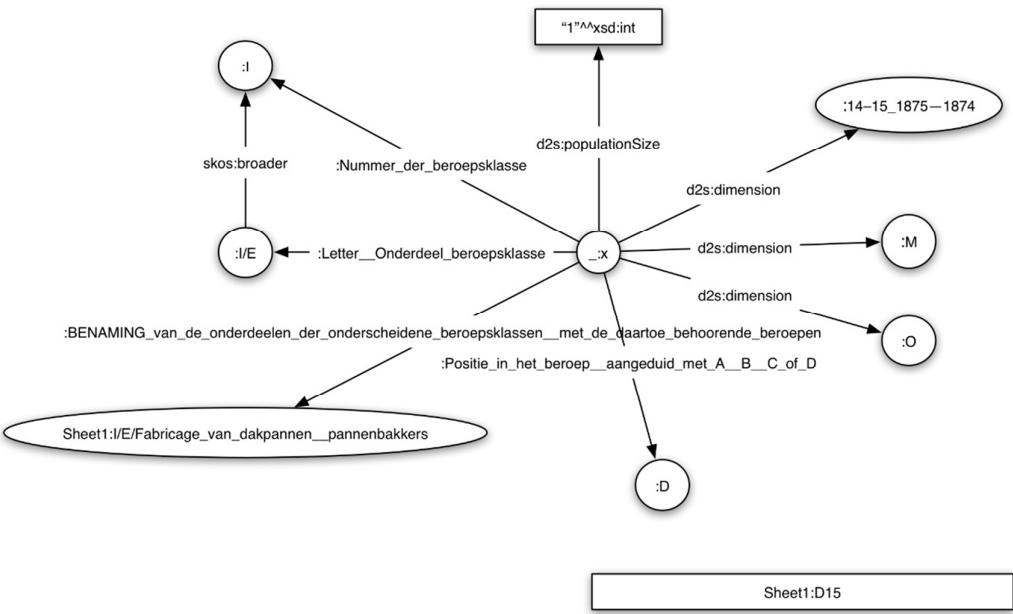
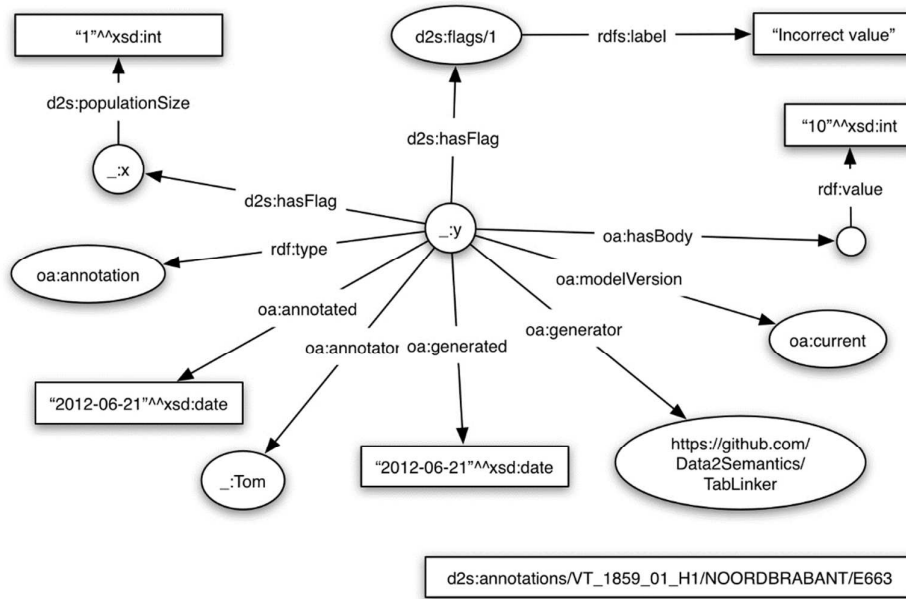


Figure 8 – Annotation graph



Year	Tables	Annotations
1795	28	100
1830	17	71
1840	60	27
1849	94	75
1859	183	4896
1869	226	321
1879	985	516
1889	166	14349
1899	76	2594
1909	138	3381
1919	4	224
1920	48	5396
1930	32	1112
1947	133	83
1956	59	138
Totals	2249	33283

Table 1 – Census Digitization statistics

Flag	Description
1	Incorrect number
2	Source error – Sum does not add up
3	Source error – Name misspelled – Corrected
4	Source error – name misspelled
5	No value
6	Number includes – Sheds
7	Not Readable

Table 2 – Annotation classification Dutch census base on a subset of the data